CAMERON BERGER

Machine Learning Engineer & Researcher ◆ (203) 942-0848 ◆ camberger203@gmail.com

ACTIVE TS/SCI

CORE SKILLS

Languages

Python, C/C++, Java, JavaScript

ML Frameworks

PyTorch, HuggingFace, LangGraph/LangChain, Scikit-learn, TensorFlow, OpenCV

Methods

Multi-Modal Deep Learning, Agentic Harnessing, Representation Learning, Reinforcement Learning, Retrieval Augmented Generation, Computer Vision, MCP Servers.

Infrastructure

Multi-GPU distributed training, HPC, Docker/Podman, Git, Weights & Biases, Linux, Windows, MacOS

Cloud

GCP, AWS, Lambda

EDUCATION

B.S. Computer Science *AI / ML Concentration*

U.S. Naval Academy · 2017–2021

CS GPA **3.93 / 4.00**

Cumulative QPR: **3.80/4.00**

COURSEWORK

Deep Learning · Intelligent Robotics · Computer Vision · Machine Learning and Data Science · High Performance Computing · Matrix Theory · Autonomy in Naval Weapons Systems

QUALIFICATIONS

Qualified Nuclear Engineer - Naval Reactors, U.S. DoE / USN · 2024

HONORS

Computer Science Department Best In-Class Project — USNA, 2021 (MovieLens Recommendation)

PROFILE

U.S. Naval Academy Computer Science graduate with research in multi-modal deep learning, reinforcement learning, genomic foundation models, and agentic AI. Five years of leadership experience as a submarine officer managing real-time control, communications systems, nuclear emergency action procedures, and reactor safety sensing and control equipment. Active TS/SCI clearance. Strong foundation in the computational science, mathematical, and systems fundamentals of modern ML, with hands-on experience with LLMs, multi-GPU training, Retrieval Augmented Generation, micro-electronic hardware, and MCP servers.

EXPERIENCE

Machine Learning Engineer: Sensors and Communication

Charles Stark Draper Laboratory, Cambridge MA. **May 2026 – Present**

- Enabling automated multi-discipline research through agentic mapping and control of sensors and device firmware.
- **DraperGPT**: Building an AI platform on Draper's internal network that gives engineers access to open-source models and tools while protecting information
- ***Retrieval-Augmented LLMs for Military Decision Support and Evaluation:*** *I/ITSEC 2620.* Co-author, developing a multi-modal decision support tool to enhance situational awareness solutions in complex high stakes environments.

Algoverse: AI Research

Genomic Foundation Model Mechanistic Interpretability **May 2026 – Sep 2026**

- ***Mechanistic Discovery in a Genomic Foundation Model Reveals a Donor-Acceptor Computation****: NeurIPS 2026, XAI4Science*. Lead author, mechanistic interpretability of a genomic foundation model to reveal learned mechanisms, and representations.

Independent Research — Multimodal Chest X-ray Anomaly Detection

Independent Self Study **Sep 2025 – Jan 2026**

- Developed an multimodal chest X-ray anomaly detection pipeline integrating MIMIC-CXR images with clinical records, laboratory results, vital signs, and clinical text. Combined MAE pretraining, ClinicalBERT embeddings, and structured patient features for pathology classification and anomaly detection.
- Average AUROC of 0.701 on 4,922 studies, with a 0.878 AUROC on edema.

Submarine Officer — Communications, IW & Reactor Controls

USS Louisiana (SSBN-743), U.S. Navy **Jun 2021 – May 2026**

- Lead survivable real-time communications, cybersecurity, and network infrastructure for a 165-person strategic nuclear submarine in GPS-denied undersea environments.
- Designed, led and taught complex training nuclear warfare simulations involving the entire ship, preparing the officer team and coordinating personnel across departments to achieve above-average performance across multiple evaluated strategic exercises.
- Lead a 12-person technical team maintaining real-time reactor control, sensor, and instrumentation systems, authored engineering assessments and delivered technical reports to Naval Reactors, U.S. Department of Energy.
- **Qualified Nuclear Engineer** by DoE and Naval Reactors after demonstrating high level of proficiency and knowledge by oral and written examination; on theory, design, and operation of Navy Nuclear Reactors.

Research Assistant — RL for Autonomous Anomaly Detection

U.S. Naval Academy **Jan 2020 – May 2021**

- Designed a deep Q-learning agent for autonomous anomaly detection in network traffic, a hybrid method combining learned policies with deterministic packet-level features.